

A Wavelet-CSP Method to Classify Hand Movement Directions in EEG based BCI System

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Abstract— The Electroencephalogram (EEG) based Brain Computer Interface (BCI) is a non-invasive system to acquire, decode and convert brain signals into control signals for an external device. The Motor-Imagery based BCI (MI-BCI) efficiently decodes the brain signals from the imagination of movement but the performance is limited by the number of commands such as right and left hand motor imageries. However, other parameters of an actual voluntary movement, such as the direction of movement, speed and extent, are encoded in the brain signals. This paper investigates the EEG brain signals from directional changes in actual hand movement. The Wavelet-Common Spatial Pattern algorithm is proposed to extract discriminative features of the brain signals that carries the direction-related information. The experiment performed on two subjects yielded a mean classification accuracy of 87.85% in decoding two classes of the direction-related information.

Keywords- Brain Computer Interface, Electroencephalogram, Discrete Wavelet Transform, Common Spatial Pattern

I. INTRODUCTION

Brain Computer Interfaces provide a non-muscular communication channel between the brain and the external world [1]. Brain signals can be acquired using invasive or non-invasive techniques and Electroencephalogram (EEG) is one of the superior non-invasive techniques. EEG is characterized by high temporal resolution, but poor spatial resolution and low signal-to-noise ratio (SNR). The high non-stationarity and low SNR nature of EEG data demand highly efficient pre-processing and feature extraction steps in order to extract the relevant information in BCI. The extracted features are often used to train a classifier to translate into control signals for an effector such as a robot or a prosthetic limb.

Motor-Imagery based BCI (MI-BCI) is one that decodes the brain signals from the imagination of left or right hand movement and is developed based on the studies that showed neural activation for movement as well as movement imaginations. The movement related features of EEG are derived from Movement Related Potential (MRP) or Event-related Desynchronization/ Synchronization (ERD/ ERS).

MRP based methods carry out time domain analysis of EEG data [2]. The neuronal activation during the preparation of movement results in contra-lateral negativation of EEG and is termed as readiness potential (RP). Several BCI systems employ features related to RP to discriminate left or right movements. MRP analysis requires simple pre-processing steps such as denoising and filtering. ERD/ERS [3] features are the commonly used neural features in EEG based BCIs. Movement preparation is accompanied by a contra lateral power decrease in mu rhythm termed as ERD. Movement and post-movement relaxation causes an increase in beta rhythm termed as ERS. The ERD/ERS analysis requires precise time and frequency patterns. The non-stationary behavior of EEG signals is a major issue in these types of analysis.

Wavelet based approach [6] is widely used for the time-frequency analysis of EEG data. The wavelet transformation provides time-frequency localization which is suited for non-stationary data analysis. In this analysis, the ERD/ERS patterns are generated by wavelet transforms and distinct features defining the task performed are extracted from these patterns. As EEG has very low spatial resolution, the BCI systems based on EEG demands efficient spatial filtering. This approach enables the subject to extract the information from the relevant brain regions. The frequently used algorithms for this are common spatial patterns (CSP) [7], independent component analysis (ICA) and various modified versions of CSP such as Common Spatio- Spectral Pattern (CSSP) [15], Filter Bank CSP (FBCSP) [12], Discriminative Filter Bank CSP (DFBCSP) [13] etc. An algorithm combining wavelet and CSP is reported in [9] in which CSP extracts the movement imagery related features from wavelet-denoised EEG data. The data set used to validate the technique in [9] was the BCI competition III dataset I in which the tasks were imagined movements of left small finger or tongue.

Several works related to BCI research has been dedicated to improving the performance of motor and motor-imagery based systems in terms of feature extraction and classification of right/left hand tasks. A major challenge in BCI research is the ability of the system to decode brain signals and interpret the various movement attributes. The hand movement kinematics include its direction, position, velocity, acceleration [11] etc.

The neuro-physiological studies affirm that such information is encoded in the neuronal signals generated from primary motor cortex region of brain. The extraction of these kinds of information can increase the number of control commands available in a BCI system. Recently, few works addressed the problem of extracting and classifying speed parameters of hand movement. Movement-related Cortical Potential (MRCP) [8] analysis has been used for this in terms of rebound rates and band powers. ERD analysis has also been performed using Event-related Spectral Perturbation (ERSP) [10] approach for the same purpose.

In this work, our objective is to derive hand movement direction-related information from brain signals. The information we aim to extract are the brain signal features associated with hand movement in different directions. This is a problem hardly addressed in literature. An ERP-feature extraction algorithm based on ICA, is proposed in [16] which analyzed the posterior parietal cortex (PPC) region during delayed saccade or reach task in different movement directions. Few recent works [4, 5] focused on finding the signal feature that carries the relevant information about movement parameters and this has lead to our hypothesis for the current experiment. The EEG signal features for hand movement in different directions are extracted and analyzed in this work. The works [4, 5] have reported that the amplitude modulation in various EEG bands carry direction information. Topographic analysis of the brain signals recorded by Electroencephalography (EEG) showed that the neuronal activation in primary motor cortex region results in these spectral modulations of brain signals [5]. Our research is based on the hypothesis that the amplitude modulation trend can also be observed in brain signal recorded by EEG but with a lower SNR. Thus the key requirements of such an analysis are high spectral resolution along with temporal specificity and spatial localization. An algorithm based on wavelet-CSP is presented in this work, which fulfills the above requirements. The method is validated using datasets collected from various subjects.

The rest of the paper is organized as follows: Section II presents the proposed Wavelet-based CSP algorithm for BCI, Section III explains the experiments conducted and detailed report on data analysis. Section IV and V presents the results and conclusion respectively.

II. WAVELET-BASED CSP ALGORITHM

The decoding of directional movement information requires the analysis of frequency spectrum of the signal in various time segments. This must be followed by spatial localization to obtain the discriminative features. FBCSP [12] is a popular method with similar objectives, which breaks down the signal to multiple frequency bands and perform CSP filtering. Spectral fine tuning needs to be performed in very low frequency range in order to obtain movement direction-specific information [5]. The fixed sub bands and the limited tuning capability of filter bank methods thus hinder such an analysis. Also the temporal variation in the non-stationary signal needs

to be tracked separately. In this scenario, a wavelet based approach can provide better performance. The Discrete Wavelet Transform (DWT) analyzes the signal at different frequency bands with different resolutions by decomposing the signal into coarse approximation and detail information [14]. DWT is a modified multi level filter bank approach. In order to spatially filter the data, the well-known CSP method is used. CSP is applied to the sub-band signals at each level, and thus the discriminative features from sub-bands are obtained. These features are passed onto a classifier based on Fisher Linear Discriminant (FLD) for further classification. FLD provides a statistical approach [12] which derives linear combination of features which characterize a particular class based on Fisher ratio.

A. Feature Extraction Algorithm

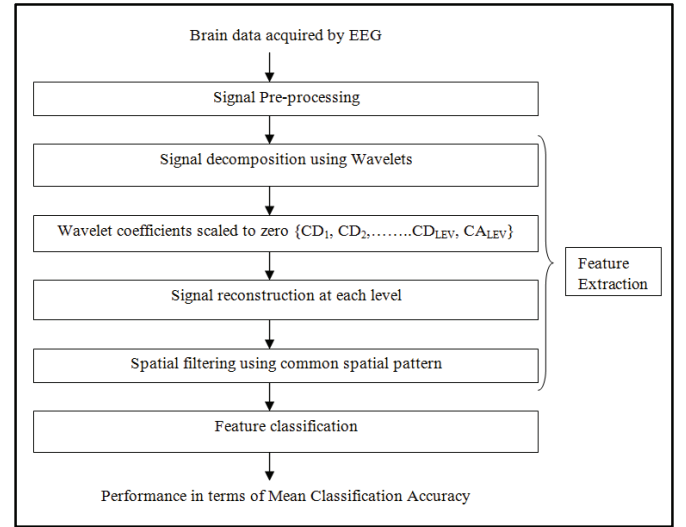


Figure 1. Proposed movement directional information decoding algorithm.

As shown in Fig. 1, the proposed method consists of four stages: data acquisition, preprocessing, feature extraction and classification. The EEG data acquired is fed to the preprocessing stage. The steps in preprocessing stage are explained in the data analysis part in Section III-B. The feature extraction stage comprises of Wavelet-CSP technique. The features extracted are given as inputs to the classifier based on FLD.

The steps involved in proposed algorithm are given below:

1. The pre-processed data is decomposed into its sub-bands using Discrete Wavelet Transform.
2. The wavelet coefficients obtained are recorded as Approximation Coefficient (CA) and Detailed Coefficients (CD).
3. The reconstruction of signals at each level is performed using the wavelet coefficients.
4. For each level, the reconstructed signal is spatially filtered using CSP technique.
5. The discriminative features obtained are gathered and is passed on to the classifier. Performance of the

system is measured in terms of mean classification accuracy.

B. Sub-band Analysis

The multi-level decomposition of the signal is performed using DWT, which employs two sets of functions called scaling and wavelet functions, which performs successive low-pass and high-pass filtering of the signal. The decomposition of the signal at each level is followed by sub sampling which halves the time resolution (as half number of samples are removed) and the frequency resolution is doubled (as the signal frequency band now spans only half the band from previous level). The wavelet decomposition structure is shown in Fig. 2.

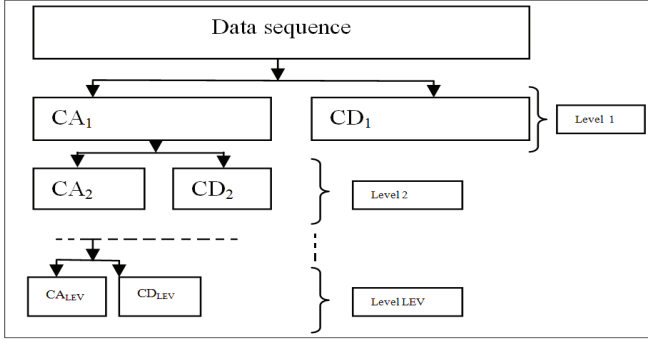


Figure 2. A LEV-level wavelet decomposition structure.

The coefficients are obtained as CA_{LEV} and CD_b , $i=1$ to LEV . The number of levels is given by $LEV = (\log_2 N)$ where N is the number of data samples. A wavelet transform scaled to zero is used for efficient analysis of very low frequency data.

The wave reconstruction is the inverse of the above mentioned process. It reconstructs the signals by using the wavelet coefficients in various sub-bands. The process involves upsampling of the sequence followed by reconstruction.

The wavelets used for signal decomposition and reconstruction are the Daubechies wavelets of order 1. These are orthogonal wavelets and are generated from an octave-band filter bank designed from two-channel orthogonal perfect reconstruction Quadrature Mirror Filter bank.

C. Common Spatial Pattern

We employ the well known CSP algorithm for computing discriminative features between two classes of actions with respect to their topographic patterns. It is based on simultaneous diagonalization of the covariance matrices of two classes. For each trial, the following operations are performed on the EEG data, E , of dimension given by Number of channel $\times$ Number of samples. The normalized spatial covariance matrix, C , of data, E is given by (1),

$$C = \frac{EE'}{\text{trace}(EE')} \quad (1)$$

The composite covariance matrix, C_c , can be regarded as the sum of covariances of the two constituent classes, (2).

$$C_c = C_a + C_b \quad (2)$$

where C_a and C_b are the covariance matrix of two classes. The composite covariance can be decomposed as (3),

$$C_c = U_c \lambda_c U_c' \quad (3)$$

using eigen vector decomposition. Here U_c is the eigen vector of C_c , U_c' is the transpose of U_c and λ_c is the diagonal matrix of eigen values. The whitening matrix, P , is calculated as shown in (4),

$$P = \sqrt{\lambda_c^{-1}} U_c \quad (4)$$

The transformed covariance matrices, S_a and S_b are obtained as (5-6):

$$S_a = P C_a P' \quad (5)$$

$$S_b = P C_b P' \quad (6)$$

The eigen vectors of both the transformed matrices are the same, given as B , and the corresponding eigen values, λ_a and λ_b add up to 1 (7-9).

$$S_b = B \lambda_b B' \quad (7)$$

$$S_a = B \lambda_a B' \quad (8)$$

$$\lambda_a + \lambda_b = I \quad (9)$$

Thus simultaneous diagonalization is carried out and the projection of whitened EEG to the smallest and largest eigen vectors are the feature vectors which optimally discriminate the two classes. The projection matrix is obtained as (10),

$$W = B'P \quad (10)$$

Thus we obtain the spatially filtered EEG, Z as (11),

$$Z = WE \quad (11)$$

The features which provide maximal discrimination are from the signals in first and last row of Z , i.e. the ones corresponding to largest and smallest eigen values. The number of such rows to be considered is given by the value m , which is typically taken as 2. Thus feature vectors are obtained as (12)

$$F_p = \log \frac{\text{var}(Z_p)}{\sum_{i=1}^{2m} \text{var}(Z_i)} \quad (12)$$

where, p varies from 1 to $2m$. The normal distribution of data is approximated using log-transformation.

III. EXPERIMENT

In this study, the experiment is performed to collect the EEG data which provides hand movement direction information. A protocol is developed to carry out hand movement in different directions. The subjects are instructed to perform actual hand movement in the specified direction. A visual feedback is provided to improve the performance of subjects.

A. Procedure

The experiment was performed using Neuroscan NuAmps 32 channel EEG amplifier. The data was recorded from the Neural Signal Processing Lab of Institute of Infocomm Research, Agency for Science, Technology and Research,

Singapore. EEG data was taken from two subjects. The task was performed using right hand and hence the data was recorded from the channels in left motor cortex. For data analysis, 12 channels around electrode C3 is used.

The hand movement directions studied are as shown in the Fig. 3. Left, Right, Forward and Backward movements correspond to West, East, North and South directions respectively.

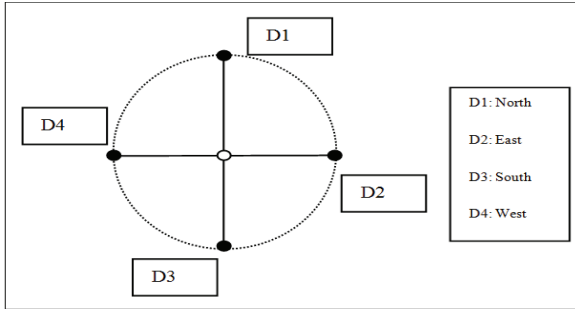


Figure 3. Four different hand directions considered in the experiment.

The timeline for the experiment protocol for a single trial is shown in Fig. 4. The visual display provided as feedback is also given.

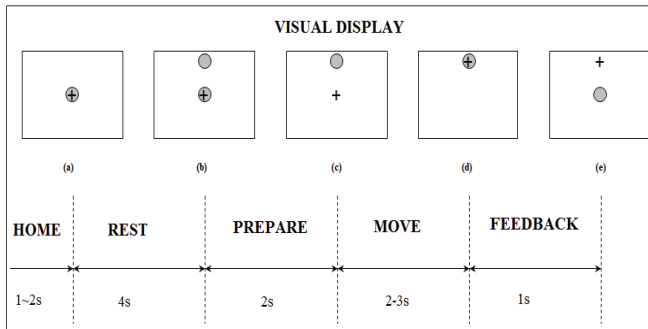


Figure 4. Experiment timeline and visual display for the task NORTH.

The subject was seated on a chair with arm resting on a table, facing a computer monitor, which provided the visual cue and feedback. Fig. 4, shows the visual provided for the task 'North', i.e., hand movement in forward direction. At the start of trial, a home screen (a) was shown with encircled cross. The target cue (circle in one of four directions) was displayed at the end of the rest period (b). The target location could be any of the four directions –North, South, East and West. At the end of preparation interval, the circle disappeared, which signals the start of task(c). The task duration was around 3 seconds, at the end of which cross appeared at the target (d). In the following 1sec. feedback for the user was provided, in which a successful trial was indicated by reappearance of the circle at the rest position (e). The display returns to home screen and this cycle was repeated.

During the task interval, subject was asked to move the right hand in a horizontal plane in the specified direction. The subject held a MIT MANUS robot in his hand while the task was performed. This device recorded the hand position at every sample time. A cycle of trials containing all 4 directions in

randomized order, was defined. Thus in each cycle, four trials of data were recorded. Each session of experiment consists of 50 cycles and 2 sessions were performed by each subject. Each trial took 10 seconds and thus the time taken to record data from each subject was around 1hour 7 minutes.

B. Data Analysis

The recorded EEG data undergoes the preprocessing steps as follows. The data was low-pass filtered at 100Hz and notch filter for 50 Hz was enabled. The sampling rate of the device was set as 250Hz. The selected EEG channels were FT7, FC3, FCz, T7, C3, Cz, TP7, CP3, CPz, P7, P3, Pz.

The data during movement preparation and execution was separately collected. The exact time samples obtained from the robot was applied in the EEG data to extract the corresponding trials. For our analysis, the data collected during movement execution alone was used. The data obtained after preprocessing was used for feature extraction. The wavelet decomposition and reconstruction at various levels were performed using Daubechies wavelets. The sub-band signal at each level was fed into a CSP, where the spatial filter corresponding to the data was generated. The feature vectors were derived as explained before. Thus for each sub-band, a set of feature vectors were obtained. The process was repeated for the entire trials.

The extracted features are used as the inputs to the classifier. In this analysis, we used the two-class Fisher Linear Discriminant classifier. In our experiment, we took two directions at a time and executed the algorithm on it. Thus we have six combinations of directions. The classification performance is evaluated using a 5 x 5 cross validation procedure. The data is divided into training and test data sets. The training dataset is used to train the classifier and the test data set is run on the classifier and the accuracy is measured as the number of trials accurately classified out of the total number of trials in test data set.

IV. RESULTS AND DISCUSSION

The average classification accuracies of the two-class classification obtained for two subjects are shown in Table I. Our algorithm is able to discriminate any two classes with utmost misclassification of 21.4% (for subject 2 in North-East classification). The mean accuracy for Subject 1 is 90.25%, Subject 2 is 85.40% and the mean classification accuracy of both subjects over all the six combinations is 87.85%.

TABLE I. AVERAGE CLASSIFICATION ACCURACY FOR 5 x 5 CROSS VALIDATION IN PERCENTAGE

Directions classified	Subject 1	Subject 2
East - West	88.36	90.94
North - South	93.26	84.23
South - East	93.60	84.50
North - East	91.24	79.60
North - West	89.03	82.68
South - West	86.02	90.46

Fig. 5 shows the CSP pattern obtained for East-West

classification. The pattern shows clear discrimination in the selected electrodes.

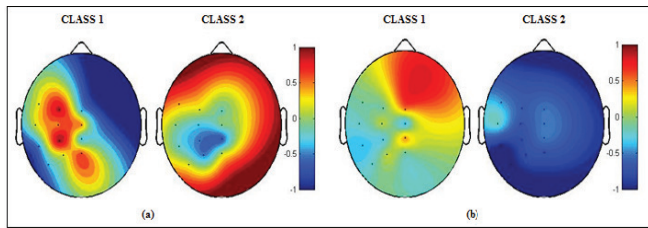


Figure 5. Spatial pattern of EEG data during East-West classification. (a) Features from level- LEV (b) Features from level-I. The plot scale is shown aside.

Our hypothesis was based on the following observations reported in [4]. The discriminative information with respect to directions is encoded in the very low (0-7 Hz) and very high (62-87 Hz) frequency bands. Hence, the sub-band signals corresponding to these spectra provide the maximum directional information. We use the spatial pattern obtained from these sub-band CSPs to validate this observation. In the current algorithm, the frequency bands mentioned correspond to the highest and lowest levels of wavelets. The CSP plot obtained justifies our hypothesis. It is observed that in the highest and lowest sub-bands the most discriminative features are shown by EEG data in the motor cortex region.

In our experiment, the main objective was to extract the information regarding the hand movement direction from EEG data. We have used the data recorded during actual movement, and hence, the primary motor cortex showed very high neuronal activity. We explored the possibility of classifying the data based on directional information in the derived features from our algorithm. The results we obtained shows that the movement parameter, direction, can be estimated from the EEG data recorded during actual movement. The algorithm in [16] analyzed PPC region using ICA and have reported an accuracy of 80.25% in binary classifier(left versus right directions). Our results show a clear improvement in performance and an effective way to address the problem of classifying hand movement directions using EEG-based BCI.

V. CONCLUSION

The prospect of BCI research lies in the rehabilitation of physically disabled patients who cannot move but can think. MI- BCIs are developed based on the trends of brain signal which were observed during actual movement. This paper investigated the presence of direction-related information on EEG data collected from actual hand movement experiment. The direction-related EEG data is extracted and classified using a two-class classifier. In order to improve the resolution in temporal, spectral and spatial domains, a Wavelet-CSP algorithm was proposed. The proposed algorithm employs a multi-level sub-band decomposition of the EEG data followed by spatial localization. The results of applying the proposed algorithm on the data collected yielded high classification accuracies of 90.25% and 85.40% in classifying the direction of hand movement, which showed that the brain signal features carry movement direction-related information.

Hence, the results obtained in this study encourage the use of new features in the brain signals to improve available control commands in EEG-based BCI. This procedure can then be further extended to motor imagery brain signal. Future work includes the extension to a multi-class paradigm in order to classify all four directions simultaneously, and further experiments that involved the analysis of motor imagery data.

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